Avi Gobrin

avichaigobrin@gmail.com | Canadian Citizen
linkedin.com/in/avigobrin | github.com/Avi-Gobrin | avi-gobrin.github.io/personal-site

Education

University of Toronto Sep 2023 - Apr 2027
Honours BSc, Mathematics & Its Applications Specialist, Statistics Major, Computer Science Minor *Toronto, ON*

Relevant Coursework: Statistical Machine Learning, Probability & Statistics, Multivariate Data Analysis, Linear Algebra, Nonlinear Optimization, Statistical Computation, Real Analysis, Mathematical Modelling, Data Structures and Algorithms

Experience

Nextdoor May 2026 - Aug 2026
Machine Learning Intern *San Francisco, CA*

- Sole engineer on Best Places for Families, an SEO ranking product scoring 350,000 US neighbourhoods on family-friendliness: owned the research question, timeline, data pipelines, model, evaluation, and user-facing page, and presented the system to the company.
- Designed the scoring methodology: screened 33 candidate metrics drawn from 10+ Census/ACS and internal tables down to 8 features, benchmarked OLS, ridge, NNLS, and monotone GBM, then hand-tuned the final weights when no fitted model matched human judgment.
- Built the evaluation harness for a problem with no ground truth: 100+ preregistered adversarial neighbourhood pairs resolved by pairwise Bradley-Terry judgments, cross-checked by LLM and manual review.
- Traced an early correlation between the score and wealth to a geographic join missing a county guard that had leaked out-of-state Census data into 94.6% of neighbourhoods; the fix removed 95% of bad links.
- Automated the whole path in Python: Airflow ingestion, a backend scoring service, and a Postgres/GraphQL path to a React page, running on Docker and Kubernetes on AWS with scheduled jobs, tests, and Datadog monitoring.
- Built Databricks tooling that lets the PM, design team, and non-technical reviewers observe any city's ranking and retune feature weights.

Shopify May 2025 - Aug 2025
Software Engineering Intern *Toronto, ON*

- Built a full-stack partner-attach system (Ruby on Rails, React, Calendly API) that screens each new merchant, matches those needing a specific language to a partner who speaks it, and assigns the rest to the least-loaded advisor, routing 100+ vetted merchants per day into guided onboarding that replaced an unassisted migration path.
- Delivered Shopify's German VAT compliance program for the entire German merchant base under a hard legal deadline: connected legacy databases to a new compliance flow, built an intake form for the data Shopify did not hold, and integrated the German tax authority's API to generate and email compliance PDFs.

Projects & Research

University of Toronto, Department of Statistical Sciences Aug 2026 - Present
Undergraduate Researcher, Advisor: Prof. Ricardo Baptista (Vector Institute) *Toronto, ON*

- Investigating a unifying rank-reduction framework linking in-context learning and fine-tuning as low-rank weight updates, connecting recent transformer interpretability work to classical multivariate statistics.
- Reducing LLM computational cost while maintaining accuracy on custom benchmarks.

Latent-Structure Modeling of Student Outcomes (OULAD) *STA437 Research Report* [Report] 2026

- Ran a full multivariate pipeline (PCA, factor analysis, Gaussian mixture clustering, LDA) over 23,301 Open University student-course registrations, recovering two dominant dimensions: assessment engagement (submission count and timing) and performance (mean score and its spread).
- Benchmarked generative against discriminative classifiers under 5-fold CV (logistic regression 67.0% vs. LDA 64.5% vs. 53.0% majority baseline); logistic regression lifted recall on the smallest class, Distinction, from 3.2% to 39.3%.

Tiny GPT From Scratch *Personal Project* [Repo] 2026

- Implemented a character-level GPT in pure NumPy with no autograd: multi-head causal attention, pre-LN Transformer blocks, Adam, and top-k sampling, with hand-written backward passes for every layer.
- Verified gradients for every model parameter against finite differences in a 56-test pytest suite, run in CI on every push.

Technical Skills

ML & Data: PyTorch, scikit-learn, NumPy, TensorFlow, pandas, SciPy, statsmodels, SQL, Airflow, Databricks
Languages & Systems: Python, R, Ruby, Java, JavaScript, Bash, AWS, Docker, Kubernetes, Datadog, Postgres, GraphQL, React, Ruby on Rails, Git, Claude Code, LLM agents
Interests: StarCraft II Grandmaster (top 100 in North America), competitive Magic: The Gathering, karate black belt